NAME:-Sheikh Wasimuddin

ROLL NO:-73

BRANCH :-CSE(IOT)

LAB 08	
Aim:	Introduction to data visualization in R.
Theory	Data visualization in R helps to present complex data in a clear and understandable way, making patterns, trends, and relationships easier to identify. It's essential for effective data analysis and communication. **Key Types of Visualizations** • **Bar Charts**: Compare categories. • **Histograms**: Show the distribution of data. • **Box Plots**: Display data spread and outliers. • **Scatter Plots**: Reveal relationships between two variables. • **Line Charts**: Show trends over time. • **Heatmaps**: Represent data in matrix form using color intensity. **Best Practices** • **Clarity**: Avoid clutter; make charts easy to interpret. • **Consistency**: Use uniform color schemes and labels. • **Right Chart**: Choose the most appropriate visualization for your data. • **Context**: Provide titles, labels, and legends for understanding. **Tools in R** • **ggplot2**: A powerful library for creating customizable static plots. • **plotly**: Allows for interactive visualizations. • **lattice**: Good for multi-panel plots. • **base R**: Simple, built-in plotting functions.
Program:	#Sheikh Wasimuddin 23071044 IOT-73 data(iris) plot(iris$Sepal.Length, iris$Sepal.Width, main="Scatter Plot of Sepal Length vs Sepal Width", xlab="Sepal Length", ylab="Sepal Width", pch=19, col="lightblue") hist(iris$Sepal.Length, main="Histogram of Sepal Length",

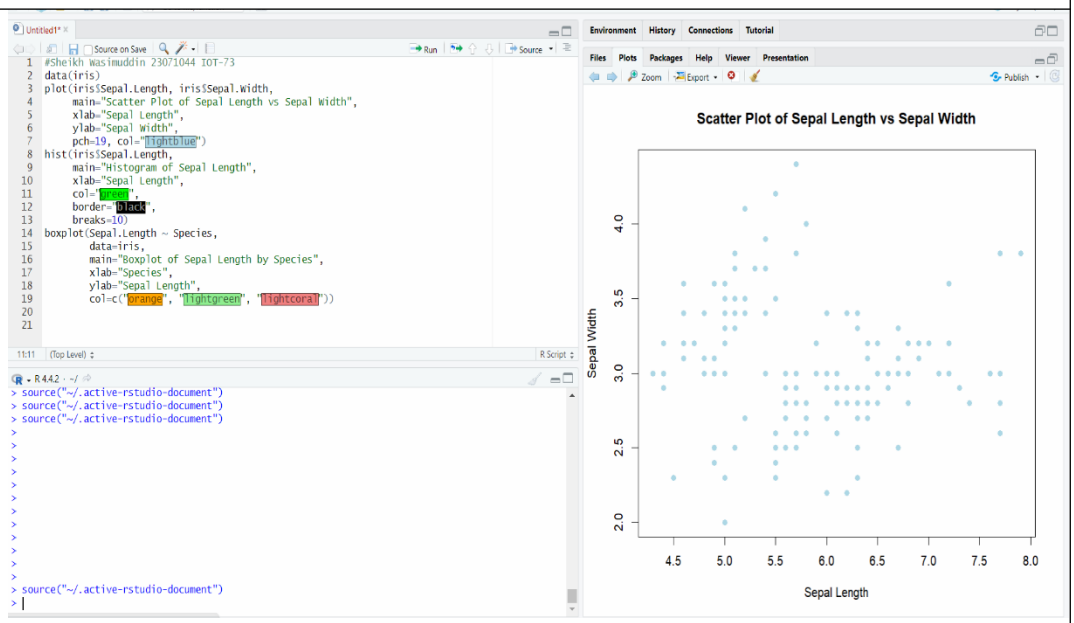
NAME:-Sheikh Wasimuddin

ROLL NO:-73

BRANCH :-CSE(IOT)

```
    xlab="Sepal Length",
    col="green",
    border="black",
    breaks=10)
boxplot(Sepal.Length ~ Species,
      data=iris,
      main="Boxplot of Sepal Length by Species",
      xlab="Species",
      ylab="Sepal Length",
      col=c("orange", "lightgreen", "lightcoral"))
```

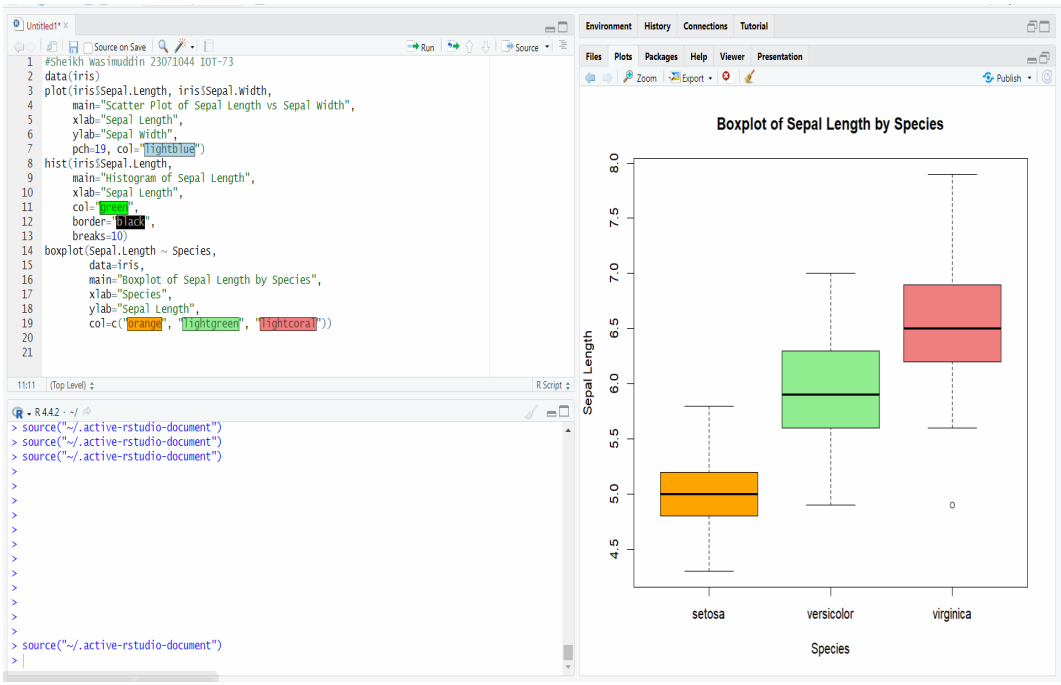
Output:



NAME:-Sheikh Wasimuddin

ROLL NO:-73

BRANCH :-CSE(IOT)



Conclusion	Introduction to data visualization in R executed successfully.
Name Of Faculty:	Prof. Akhil Jajulwar